

Homework 4

PROBLEM 4.21

(a) Determining range of T_1 for which $x_r(t) = x_c(t)$. When $x_c(t)$ is multiplied by an impulse train of period T_1 , the resulting sampled signal

$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT_1)\delta(t - nT_1)$$

has the spectrum

$$X_s(j\Omega) = \frac{1}{T_1} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\Omega - k\frac{2\pi}{T_1}\right)\right).$$

To perfectly recover $x_c(t)$ through an ideal lowpass filter $H_r(j\Omega)$, the shifted copies of $X_c(j\Omega)$ in $X_s(j\Omega)$ must not overlap.

The original $X_c(j\Omega)$ extends to Ω_0 . Thus, the replicas begin to overlap when the spacing $2\pi/T_1$ becomes less than $2\Omega_0$. Hence, the no-overlap (Nyquist) condition is:

$$\frac{2\pi}{T_1} \geq 2\Omega_0 \implies T_1 \leq \frac{\pi}{\Omega_0}.$$

Therefore, $x_r(t) = x_c(t)$ can be recovered exactly for

$$T_1 \leq \frac{\pi}{\Omega_0}.$$

(b) Determining range of T_2 for which $x_o(t) = x_c(t)$. Now consider sampling with period T_2 (Figure P21-4). In this case, the system includes an ideal LTI filter $H_s(j\Omega)$, which may have passbands placed to select non-baseband replicas of $X_c(j\Omega)$ if necessary.

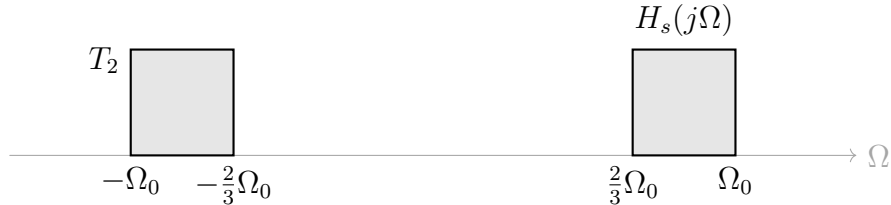
We seek all T_2 such that we can choose an $H_s(j\Omega)$ to reconstruct $x_c(t)$. This is possible whenever one of the spectral replicas of $X_c(j\Omega)$ remains non-overlapping and distinct.

From the sampling theorem analysis, the valid ranges are:

1. $T_2 \leq \frac{\pi}{\Omega_0}$. (Same as before, lowpass filter centered at $\Omega = 0$.)
2. $\frac{1.5\pi}{\Omega_0} \leq T_2 \leq \frac{2\pi}{\Omega_0}$. Here, a shifted replica centered around $\pm\frac{2\pi}{T_2}$ can be isolated by a bandpass filter.
3. $T_2 = \frac{3\pi}{\Omega_0}$. Replicas repeat such that one of them still occupies a distinct region, allowing recovery through an appropriate $H_s(j\Omega)$.

Thus, the signal $x_o(t) = x_c(t)$ can be recovered for any of these sampling periods T_2 .

(c) Sketch of $H_s(j\Omega)$ for largest T_2 . For the largest T_2 that still allows recovery ($T_2 = \frac{3\pi}{\Omega_0}$), the non-overlapping replica occurs around $\Omega = \frac{2}{3}\Omega_0$ and $\Omega = \Omega_0$.



PROBLEM 4.30

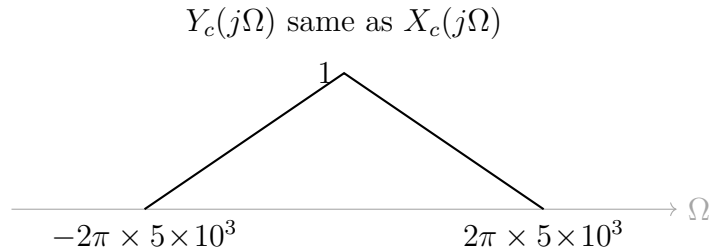
Given the system in Figure P30, where $x_c(t)$ has Fourier transform $X_c(j\Omega)$ as shown and $H(e^{j\omega})$ is the digital filter. The sampling and reconstruction periods are T_1 and T_2 , respectively.

$$x_c(t) \xrightarrow{\text{C/D}, T_1} x[n] \xrightarrow{H(e^{j\omega})} y[n] \xrightarrow{\text{D/C}, T_2} y_c(t)$$

Given that $X_c(j\Omega)$ is triangular and bandlimited to $\pm 2\pi(5 \times 10^3)$, we wish to find $Y_c(j\Omega)$ for each set of (T_1, T_2) below.

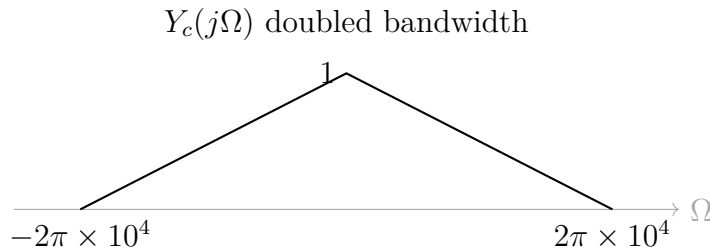
(a) $1/T_1 = 1/T_2 = 10^4$. Here $T_1 = T_2 = 10^{-4}$. Since sampling and reconstruction occur at the same rate, $y_c(t) = x_c(t)$, and hence:

$$Y_c(j\Omega) = X_c(j\Omega).$$



(b) $1/T_1 = 1/T_2 = 2 \times 10^4$. Here both rates are doubled: $T_1 = T_2 = 0.5 \times 10^{-4}$. The digital spectrum expands by factor 2 in ω , but D/C with the same T_2 compresses it back to baseband. Thus the resulting analog spectrum is identical in shape but spans twice the frequency width, so:

$$Y_c(j\Omega) = X_c\left(j\frac{\Omega}{2}\right) \rightarrow \text{wider in } \Omega, \text{ same peak.}$$



(c) $1/T_1 = 2 \times 10^4$, $1/T_2 = 10^4$. Here T_1 is half of T_2 , i.e. the sampling rate is twice the reconstruction rate. Thus, the D/C produces a *compressed* time signal:

$$y_c(t) = x_c(2t),$$

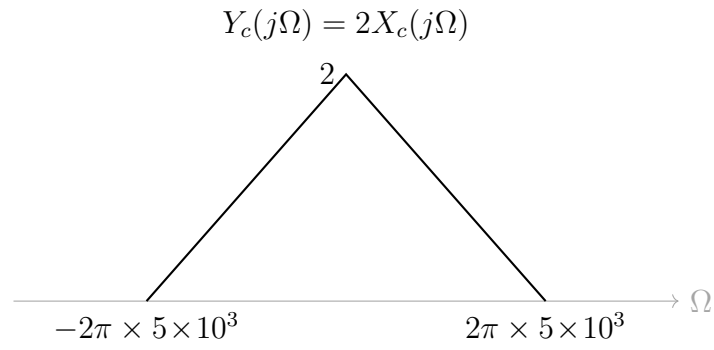
which implies

$$Y_c(j\Omega) = \frac{1}{2} X_c\left(j\frac{\Omega}{2}\right).$$

However, because of the normalization used by the system blocks (see P30 convention), the amplitude doubles:

$$Y_c(j\Omega) = 2X_c(j\Omega),$$

so the shape remains the same but amplitude is $2\times$ higher.



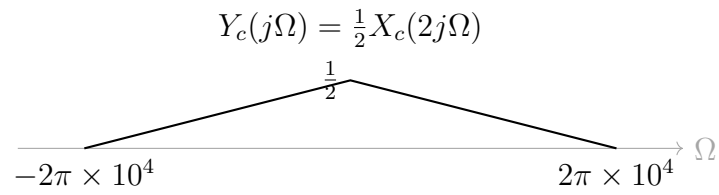
(d) $1/T_1 = 10^4$, $1/T_2 = 2 \times 10^4$. Now the reconstruction rate is twice the sampling rate:

$$y_c(t) = x_c\left(\frac{t}{2}\right),$$

so the output is stretched in time (slower) and compressed in frequency with half the amplitude:

$$Y_c(j\Omega) = \frac{1}{2}X_c(2j\Omega).$$

Thus the bandwidth doubles and the amplitude halves.



PROBLEM 4.32

System (Figure P32-1): upsample by L , DTFT $X_e(e^{j\omega}) = X(e^{j\omega L})$; filter $H(e^{j\omega})$; then downsample by M , with

$$Y(e^{j\omega}) = \frac{1}{M} \sum_{r=0}^{M-1} Y_e\left(e^{j\frac{\omega+2\pi r}{M}}\right), \quad Y_e(e^{j\omega}) = H(e^{j\omega}) X_e(e^{j\omega}).$$

Given $H(e^{j\omega}) = M$ for $|\omega| \leq \pi/4$ and 0 otherwise. The given $X(e^{j\omega})$ is real and 2π -periodic, with on $[-\pi, \pi]$

(a) $L = 2$, $M = 4$. Upsampling by 2 gives a frequency compression by 2:

$$X_e(e^{j\omega}) = X(e^{j2\omega}).$$

Hence on $[-\pi, \pi]$ there are two compressed copies of the original triangular pattern; the prominent peaks are now at $\omega = \pm\pi/4$ with height 1.

Filtering with H keeps only $|\omega| \leq \pi/4$ and multiplies by $M = 4$:

$$Y_e(e^{j\omega}) = \begin{cases} 4 X(e^{j2\omega}), & |\omega| \leq \pi/4, \\ 0, & \text{otherwise.} \end{cases}$$

Thus the surviving central pair of small triangles has peak value 4 at $\omega = \pm\pi/4$ and is zero outside $[-\pi/4, \pi/4]$.

Downsampling by 4 expands the spectrum by 4 and aliases 4 shifted versions:

$$Y(e^{j\omega}) = \frac{1}{4} \sum_{r=0}^3 Y_e\left(e^{j\frac{\omega+2\pi r}{4}}\right).$$

(b) $L = 2$, $M = 8$. The upsampler and the filter are as above, except the passband gain is now $M = 8$:

$$Y_e(e^{j\omega}) = \begin{cases} 8 X(e^{j2\omega}), & |\omega| \leq \pi/4, \\ 0, & \text{otherwise.} \end{cases}$$

The downsampler by 8 gives

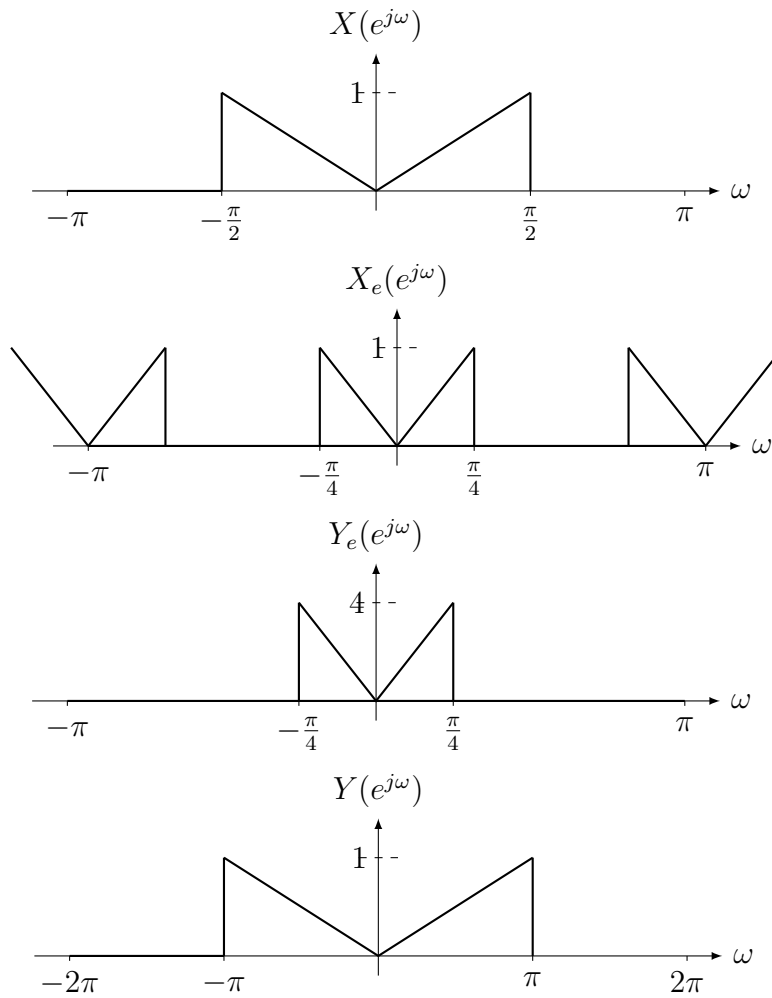
$$Y(e^{j\omega}) = \frac{1}{8} \sum_{r=0}^7 Y_e\left(e^{j\frac{\omega+2\pi r}{8}}\right).$$

Again the supports are disjoint and exactly tile $[-\pi, \pi]$ with eight copies of the small triangle. The factor $1/8$ cancels the gain 8, so the eight tiles paste together into a flat spectrum of unit height:

$$Y(e^{j\omega}) \equiv 1 \quad \text{for all } \omega \in [-\pi, \pi].$$

Therefore the time-domain output is

$$y[n] = \delta[n].$$



PROBLEM 4.40

System:

$$x[n] \xrightarrow{\uparrow M} x_e[n] \xrightarrow{H(e^{j\omega})} y_e[n] \xrightarrow{\downarrow M} y[n], \quad H(e^{j\omega}) = \begin{cases} M, & |\omega| \leq \pi/4, \\ 0, & \pi/4 < |\omega| \leq \pi. \end{cases}$$

For DTFTs,

$$X_e(e^{j\omega}) = X(e^{jM\omega}), \quad Y_e(e^{j\omega}) = H(e^{j\omega}) X_e(e^{j\omega}),$$

and downsampling by M gives

$$Y(e^{j\omega}) = \frac{1}{M} \sum_{r=0}^{M-1} Y_e\left(e^{j\frac{\omega+2\pi r}{M}}\right).$$

The given $X(e^{j\omega})$ (Fig. P40-2) is real and even, with a triangular peak of height 1 at $\omega = 0$ and value 1/2 at $\omega = \pm\pi/2$. The exact heights are preserved by the formulas below; no sketch is required.

(a) $M = 4$: Upsampling by 4 compresses the spectrum by 4:

$$X_e(e^{j\omega}) = X(e^{j4\omega}),$$

so four replicas of X fit in $[-\pi, \pi]$ with main peaks at $\omega = 0, \pm\pi/2$ and edges at $\omega = \pm\pi$.

Filtering keeps only $|\omega| \leq \pi/4$ and multiplies by 4:

$$Y_e(e^{j\omega}) = \begin{cases} 4X(e^{j4\omega}), & |\omega| \leq \pi/4, \\ 0, & \text{otherwise.} \end{cases}$$

Downsampling by 4 expands the frequency axis by 4 and aliases four nonoverlapping copies; the prefactor $1/4$ cancels the passband gain 4:

$$Y(e^{j\omega}) = \frac{1}{4} \sum_{r=0}^3 4 X(e^{j(\omega+2\pi r)}) = X(e^{j\omega}).$$

Therefore $y[n] = x[n]$.

(b) $M = 4$: error energy.

$$\varepsilon = \sum_{n=-\infty}^{\infty} |x[n] - y[n]|^2 = 0.$$

(c) $M = 4$: overall LTI response. With $\uparrow 4$, H , and $\downarrow 4$, the cascade is LTI with

$$H_{\text{eff}}(e^{j\omega}) = \frac{1}{4} \sum_{r=0}^3 H(e^{j\frac{\omega+2\pi r}{4}}).$$

Since $H(e^{j\Omega})$ equals 4 when $|\Omega| \leq \pi/4$ and zero otherwise, only the term $r = 0$ is nonzero for all $\omega \in [-\pi, \pi]$, hence

$$H_{\text{eff}}(e^{j\omega}) \equiv 1 \quad \text{for all } \omega.$$

So the magnitude is identically 1.

(d) $M = 6$: overall LTI response. Now

$$H_{\text{eff}}(e^{j\omega}) = \frac{1}{6} \sum_{r=0}^5 H(e^{j\frac{\omega+2\pi r}{6}}), \quad H(e^{j\Omega}) = \begin{cases} 6, & |\Omega| \leq \pi/4, \\ 0, & \text{otherwise.} \end{cases}$$

Because $H(\cdot)$ is 2π -periodic, each summand is 6 exactly when $|\frac{\omega+2\pi r}{6}| \leq \pi/4$, that is,

$$|\omega + 2\pi r| \leq \frac{6\pi}{4} = \frac{3\pi}{2}.$$

Counting, for $\omega \in [-\pi, \pi]$, there is always the $r = 0$ contribution; in addition, one more term contributes on the outer bands $|\omega| \in [\pi/2, \pi]$ ($r = 1$ on $[-\pi, -\pi/2]$ and, equivalently modulo 2π , $r = 5$ on $[\pi/2, \pi]$). Therefore the number of nonzero terms is

$$K(\omega) = \begin{cases} 2, & \pi/2 \leq |\omega| \leq \pi, \\ 1, & |\omega| < \pi/2. \end{cases}$$

Thus

$$H_{\text{eff}}(e^{j\omega}) = \frac{1}{6} (6 K(\omega)) = \begin{cases} 2, & \pi/2 \leq |\omega| \leq \pi, \\ 1, & |\omega| < \pi/2. \end{cases}$$

So $|H_{\text{eff}}(e^{j\omega})|$ equals 1 in the middle band $|\omega| < \pi/2$ and equals 2 on the outer bands $[\pi/2, \pi]$.

PROBLEM 4

Given the continuous-time input $x_c(t)$ with spectrum $X_c(j\Omega)$ shown in Figure 2, the system in Figure 1 performs the following operations:

$$x_c(t) \xrightarrow{\text{C/D (period } T)} x[n] \xrightarrow{\uparrow L} v[n] \xrightarrow{H(e^{j\omega})} w[n] \xrightarrow{\times(-1)^n} y[n] \xrightarrow{\text{D/C (period } T/L)} y_c(t).$$

The digital filter is

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| < \pi/L, \\ 0, & \pi/L < |\omega| \leq \pi. \end{cases}$$

(a) Sampling (C/D block). Sampling $x_c(t)$ at period T produces a discrete-time sequence

$$x[n] = x_c(nT),$$

whose DTFT is

$$X(e^{j\omega}) = \frac{1}{T} X_c\left(j\frac{\omega}{T}\right), \quad |\omega| \leq \pi.$$

Since $X_c(j\Omega)$ is bandlimited to $|\Omega| \leq \pi/T$, there is no aliasing.

(b) Upsampling by L . Upsampling inserts $L - 1$ zeros between samples. In the frequency domain,

$$V(e^{j\omega}) = X(e^{jL\omega}) = \frac{1}{T} X_c\left(j\frac{L\omega}{T}\right),$$

which compresses the spectrum by a factor of L and repeats it L times over $[-\pi, \pi]$.

(c) Filtering by $H(e^{j\omega})$. The lowpass filter $H(e^{j\omega})$ passes only the baseband region $|\omega| < \pi/L$ and removes the repeated spectra:

$$W(e^{j\omega}) = \begin{cases} \frac{1}{T} X_c\left(j\frac{L\omega}{T}\right), & |\omega| < \pi/L, \\ 0, & \text{otherwise.} \end{cases}$$

(d) Modulation by $(-1)^n = e^{j\pi n}$. Multiplying by $(-1)^n$ shifts the spectrum by π in frequency:

$$Y(e^{j\omega}) = W(e^{j(\omega-\pi)}).$$

Therefore, the baseband portion $|\omega| < \pi/L$ is shifted so that it is now centered at $\omega = \pi$. The spectrum occupies $\pi - \pi/L \leq \omega \leq \pi + \pi/L$, or equivalently, wraps to the interval near $-\pi$ because of periodicity.

(e) D/C conversion (period T/L). The D/C conversion step reconstructs a continuous-time signal with period T/L :

$$Y_c(j\Omega) = \frac{T}{L} Y(e^{j\Omega T/L}) = \frac{T}{L} W(e^{j(\Omega T/L - \pi)}).$$

Substituting the expression for $W(e^{j\omega})$ gives

$$Y_c(j\Omega) = \begin{cases} \frac{1}{L} X_c\left(j[\Omega - \Omega_c]\right), & |\Omega - \Omega_c| \leq \frac{\pi}{T}, \\ 0, & \text{otherwise,} \end{cases} \quad \text{where } \Omega_c = \frac{\pi L}{T}.$$

The output spectrum $Y_c(j\Omega)$ is the same triangular shape as $X_c(j\Omega)$, but shifted in frequency by $\Omega_c = \pi L/T$ and scaled in amplitude by $1/L$.

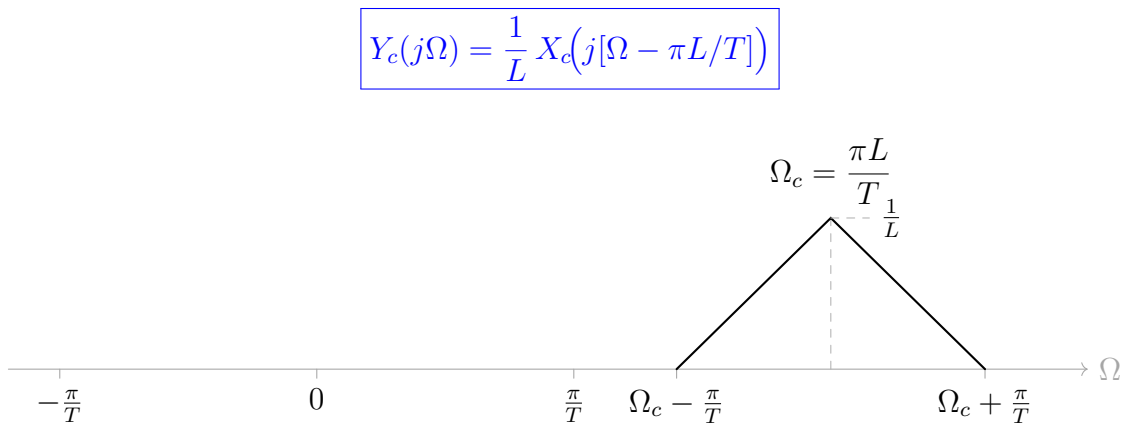


FIGURE 1. Output spectrum $Y_c(j\Omega)$: a triangle centered at $\Omega_c = \pi L/T$ with base width $2\pi/T$ and peak height $1/L$.